



Shangari-La Hotel Database System

**Diploma in Software Engineering
Data Management 1 — Course Work Proposal
DSE26.1F**

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Data Management 1 — Course Work Proposal

Database Design and Development for a Selected Organization

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Abstract

This document outlines the proposal for designing and developing a database for Shangri-la Hotel, a hotel business located in Colombo, Sri Lanka. It has been created as part of the Data Management 1 Course Work for the Diploma in Software Engineering (DSE26.1F) at the National Institute of Business Management (NIBM).

The hotel currently manages important areas including guests, rooms, room types, bookings, payments, services, employees, suppliers, guest reviews and maintenance records. Without a database system information in these areas can become disorganized. This may lead to problems such as entries, conflicts in room bookings incorrect payment records, difficulty in tracking guest details and inaccurate reports.

To solve this the proposed solution follows the database design life cycle required by the coursework. First is analysis, where the needs of Shangri-la Hotel are studied. Then comes conceptual design, which includes an ER diagram with ten entities. The diagram also shows cardinality ratios, participation constraints and assumptions made during the design process.

Next is design, where the ER model is converted into a relational schema. This involves identifying keys and foreign keys. The tables are then normalized to achieve Third Normal Form to reduce redundancy and improve data integrity.

After that table design is done, specifying the columns, data types and field sizes for each table. Finally physical design takes place using Microsoft SQL Server. This includes setting up constraints inserting sample data—minimum of ten records per table—and writing queries to demonstrate functionality.

Five basic SELECT queries are included to show retrieval of data. Five advanced queries use GROUP BY, ORDER and HAVING clauses to analyze grouped data and apply conditions. Three JOIN queries demonstrate how related data, from tables can be combined effectively.

Security is also considered with five user roles, each assigned specific permissions to ensure data access is controlled and secure.

Keywords: Database Design, ER Diagram, Relational Schema, Normalization, SQL Server, Hotel Management System, Primary Key, Foreign Key, Data Management 1 NIBM, Shangri-la Hotel.

Acknowledgement

This document presents the proposal for the design and development of a database for Shangri-la Hotel, a hotel business located in Colombo, Sri Lanka. It is prepared as the Data Management 1 Course Work of the Diploma in Software Engineering (DSE26.1F) at the National Institute of Business Management (NIBM).

Shangri-la Hotel must manage guests, rooms, room types, bookings, payments, services, employees, suppliers, reviews and maintenance records. Without a database these details can become scattered. This scattering may cause records, room booking conflicts, payment errors, difficulty in tracking guest information and unreliable reports.

The proposed solution follows the full database design life cycle required for the coursework. It includes requirement analysis; design with an ER diagram containing 10 entities, cardinality ratios, participation constraints and assumptions; logical design that maps the ER diagram to a relational schema with primary and foreign keys and normalizes the schema to Third Normal Form; table design that defines columns, data types and sizes; and physical design in Microsoft SQL Server. The physical design contains constraints, sample data with least ten records per table five basic SELECT queries, five advanced queries that use GROUP BY ORDER BY and HAVING three JOIN queries and five user levels with different permissions.

Keywords: Database Design, ER Diagram, Relational Schema, Normalization, SQL Server, Hotel Management System, Primary Key, Foreign Key, Data Management 1 NIBM, Shangri-la Hotel.

We would like to express our gratitude to everyone who supported us throughout this coursework.

First, we would like to thank the lecturers and staff of the School of Computing and Engineering at the National Institute of Business Management (NIBM). They provided guidance and a structured framework for the Data Management 1 module. This module helped us develop our knowledge of ER modelling, relational schema mapping, normalization and SQL.

We would also like to thank the management of Shangri-la Hotel for providing information about the hotels business processes. These processes include guest management, room bookings, payments, services and other hotel operations. This information helped us identify the requirements for developing the Shangri-la Hotel Database Management System.

Finally, we would like to thank our group members for their teamwork, commitment and valuable contributions throughout the project. We are also grateful to our families and friends, for their support and encouragement.

This project represents our effort to apply the knowledge and skills we have learned in Data Management 1 to a real-world hotel database management problem.

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List of Acronyms and Abbreviations

Acronym	Full Form
SHDS	Shangri-la Hotel Database System
NIBM	National Institute of Business Management
DSE	Diploma in Software Engineering
DBMS	Database Management System
ER / ERD	Entity Relationship / Entity Relationship Diagram
PK	Primary Key
FK	Foreign Key
SQL	Structured Query Language
T-SQL	Transact-SQL (Microsoft SQL Server dialect)
DDL	Data Definition Language
DML	Data Manipulation Language
DCL	Data Control Language
UNF	Unnormalized Form
1NF / 2NF / 3NF	First / Second / Third Normal Form
FAQ	Frequently Asked Questions
LMS	Learning Management System
SOCE	School of Computing and Engineering

Chapter 1: Introduction

1.1 Introduction of the Organization

Shangri-la Hotel is a hotel business located in Colombo, Sri Lanka. Shangri-la Hotel offers accommodation and hospitality services to guests. Shangri-la Hotel lets guests make room reservations, choose room types, check in and check out and pay using available payment methods. Shangri-la Hotel also offers services to guests during their stay. Behind these operations Shangri-la Hotel manages room availability, guest information, bookings, payments, employees, suppliers and maintenance activities. Each of these activities generates data. A structured relational database is the way to store and manage this data so that it stays consistent, secure and easy to access, update and query. The proposed Shangri-la Hotel Database Management System will help organize Shangri-la Hotels operations, reduce duplicate records, prevent booking conflicts and improve the efficiency of Shangri-la Hotel management.

1.2 Organization Structure and Users

Shangri-la Hotel is a hotel organization where one person can take on roles. Because of this the database focuses on roles instead of a strict staff hierarchy. It is expected that the roles listed below could be handled by the individual depending on the needs and size of the hotel.

Level	Role	Responsibility in the Hotel	Database Access
1	Owner / Administrator	Manages the hotel and the database	Full control
2	Reception and Booking Staff	Handles guests, room bookings and payments	Read / Write on booking data
3	Inventory and Procurement Staff	Manages hotel supplies, inventory and suppliers	Read / Write on inventory data
4	Service and Maintenance Staff	Manages hotel services, room maintenance and related activities	Read / Write on service and maintenance data
5	Management Reporting	Views hotel performance and management reports	Read-only

1.3 Current Operations

In a hotel that does not have a database, guest details, room information, bookings, payment records, service details and maintenance records might be kept in different files, messages or notes. The same data could be entered times. Updates might happen in one place but not in others. This makes it hard to bring all the information together when creating reports. Such issues can lead to problems, like bookings, wrong payment records trouble checking if rooms are available and reports that do not give accurate results. A centralized Sangarilla Hotel Database Management System can store all this information in one place. It helps make managing, updating and getting data much easier.

1.4 Problem Definition

The lack of a database leads to several issues:

Problem 1. Data redundancy and inconsistency: The same guest, room or booking information might be saved in places. These records may not match, causing confusion.

Problem 2. Room and booking mismatch: There is no link between room availability and actual bookings. This means the same room can end up being booked more than once by mistake.

Problem 3. Booking and payment tracking: To find a booking, payment or guest record staff must check different sources. This takes time. Slows down operations.

Problem 4. Reporting and security: Creating reports about the hotel's performance is hard. Also, people who should not have access might. Change sensitive data.

1.5 Project Objectives

- To find the entities, attributes and relationships of Sangarilla Hotel using at least 8 entities.
- To build the model as an ER diagram showing cardinality ratios, participation constraints and assumptions.
- To convert the ER diagram into a schema using standard mapping steps and find primary keys and foreign keys.
- To make the tables go up to Third Normal Form (3NF) and describe columns, data types and sizes.
- To set up the database in Microsoft SQL Server with keys, constraints and at least 10 records, per table.
- To answer hotel business questions using SELECT, GROUP BY, ORDER BY HAVING and JOIN queries.
- To make five levels of database users. Give each level the right database permissions.

1.6 Proposed Solution

A relational database called Shangri-la Hotel Database System (SHDS) will be built and run using Microsoft SQL Server. The database will include 12 tables in total—10 entities and 2 additional tables created during the mapping process. To keep the data accurate and consistent keys, foreign keys and constraints such as CHECK, UNIQUE, NOT NULL and DEFAULT will be applied. Access to each table will be managed through five user levels, which determine who can view or change the data.

1.7 Scope of the Project

The proposed Hotel Management System will cover the main day-to-day operations of a hotel. It will manage guest registration and profile details, room categories, room availability, reservations, and the check-in and check-out process. The system will also handle billing and payment records, including any outstanding balances for guests.

Employee and department details will be maintained within the system, while housekeeping activities can be recorded and assigned to employees. Room service requests made by guests will also be recorded so that the hotel can keep track of the services requested.

The system will include restaurant management, where restaurant tables and customer orders can be recorded. Supplier details and hotel inventory will also be managed to keep track of supplies used by the hotel. In addition, the system will support event and conference bookings by storing the relevant booking details.

Guest feedback will also be recorded and maintained for future reference. Finally, SQL queries will be used to generate useful information from the database, such as room occupancy status, pending reservations, billing summaries, employee task loads, inventory information, and popular dining items.

In Scope	Out of Scope
Requirement analysis of Shangri-la Hotel Conceptual design which includes an ER diagram design which includes a relational schema and normalization Table design that includes data types and sizes Physical design, in SQL Server which includes DDL, DML and DCL SQL queries and security design	A website or mobile application interface An online payment gateway integration An integration, with an external courier system Data warehousing An analytics dashboard

1.8 Chapter Summary

I found this chapter introduces Shangri-la Hotel, the people who use it, the problems that come from not having a database, the goals of the project, the solution that was suggested and the limits of the project. Chapter 2 looks at the data needs of Shangri-la Hotel.

Chapter 2: Analysis

2.1 Introduction

This chapter explains how the requirements are gathered and what data must be stored for Shangri-la Hotel. It identifies the entities, attributes, business rules and assumptions and lists the asked questions (FAQs) that the database should answer.

2.2 Requirement Gathering Methods

Method	Description	Outcome
Discussion with the business	Understand guest, room booking, payment and service processes	List of entities and business rules
Review of business records	Study sample guest, room, booking and payment records	Attributes and data formats
Study of hotel operations	Examine room types, reservations, check-in check-out and services	Guest, room and booking data
Process walkthrough	Follow a booking from reservation to check-out and payment	Process flow and requirements
Relationships and cardinality	Identify relationships, between guests, rooms, bookings, payments and other entities	Relationships and cardinality

2.3 Functional Requirements

1. The system shall register and store guest profile information including name, contact details, NIC, address, and nationality.
 2. The system shall record and manage room details including room number, floor, status, and room type category.
 3. The system shall store room type information including type name, description, base price, and maximum occupancy.
 4. The system shall create, update, and track guest reservations including check-in date, check-out date, number of guests, and reservation status.
 5. The system shall record payment transactions linked to reservations, including amount, payment date, payment method, and payment status.
 6. The system shall store and manage hotel staff records including name, email, phone, role, hire date, and department assignment.
 7. The system shall maintain department records including department name and the manager responsible for each department.
 8. The system shall record service types offered by the hotel including service name, description, and price.
 9. The system shall record service orders placed by guests including the service requested, order date, quantity, and total cost.
 10. The system shall store supplier information including company name, contact person, email, phone, and product category.
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11. The system shall support SQL queries to retrieve room availability and occupancy status.
12. The system shall support SQL queries to retrieve pending reservations and billing summaries.
13. The system shall support advanced SQL queries using GROUP BY, ORDER BY, and HAVING clauses for operational reporting.
14. The system shall support JOIN queries to retrieve combined data across related tables.
15. The system shall create different user roles and grant controlled permissions to each role based on their responsibilities.

2.4 Non-Functional Requirements

1. Performance — The database shall retrieve query results within an acceptable response time even when all tables contain multiple records simultaneously.
2. Reliability — The system shall maintain data integrity at all times through the use of primary keys, foreign keys, and constraints on all tables.
3. Security — Access to the database shall be controlled through role-based user accounts, ensuring staff can only access data relevant to their department and responsibilities.
4. Data Integrity — All tables shall enforce referential integrity so that no orphan records exist across related tables.
5. Scalability — The database structure shall be capable of accommodating a growing number of guests, reservations, staff records, and service orders without requiring structural redesign.
6. Maintainability — The SQL script shall follow consistent naming standards and include appropriate comments so that the database can be understood, updated, and maintained by any developer.
7. Usability — The database shall be implemented in Microsoft SQL Server, a widely used and well-supported platform, ensuring ease of access and management for authorized users.
8. Availability — The database shall be structured so that authorized users can access operational data at any time without dependency on manual processes or external systems.
9. Consistency — Data entered across all tables shall follow consistent formats and naming conventions to prevent duplication and confusion.
10. Normalization — All tables shall satisfy Third Normal Form to eliminate data redundancy and ensure clean, non-redundant data storage.

2.5 Entities and Attributes

Ten entities were identified for the Shangri-La Hotel Database System (SHDS) meeting the coursework requirement of least eight entities. The key attributes are underlined in the design presented in Chapter 3.

Entity	Primary Key	Attributes
GUEST	<u>GuestID</u>	FirstName, LastName, Email, Address, NIC, Nationality, <u>Phone</u>

Entity	Primary Key	Attributes
RoomType	RoomTypeID	TypeName, Description, BasePrice, MaxOccupancy
Room	RoomID	RoomNumber, Floor, status
Reservation	ReservationID	CheckInDate, CheckOutDate, NumGuests, Status
Payment	PaymentID	PaymentMethod, PaymentDate, Amount, Status
Staff	StaffID	FirstName, LastName, Email, Phone, Role, HireDate
Department	DepartmentID	ManagerID, DepartmentName
Service	ServiceID	ServiceName, Description, ServiceName, Price
ServiceOrder	OrderID	OrderDate, Quantity, TotalCost
Supplier	SupplierID	CompanyName, Phone, ProductCategory, Email, ContactPerson

2.6 Attribute Types

Type	Attribute	Transformation
Composite	Guest name (first_name, last_name); Address (street, city district)	Stored as atomic columns
Multivalued	Guest phone_no	Moved to separate GUEST_PHONE table
Derived	Booking total_amount = SUM(quantity × unit_price)	Calculated by query not stored

2.7 Business Rules and Assumptions

- A guest can make a booking. Can register without making a booking. Every booking is linked to one guest.
 - A room is assigned to one room type. A room type can exist before any room is added to it.
 - A booking can include one or more rooms. A single room can be part of bookings over time.
 - The BOOKING_ROOM table stores the room price at the time the booking was made. This ensures that any future changes to room prices do not impact bookings.
 - A booking may have one or more payment records. For example a payment might fail first. Then succeed later.
 - If a booking includes hotel services it will have related service records. If no extra services are added then no service records exist.
-

- An employee can process bookings. An employee can have at one manager. This is a relationship.
- A guest can submit a review for a hotel service or for a stay. The review includes a rating from 1 to 5.
- The sample data used in the database is, for illustration only.

2.8 Chapter Summary

Ten entities, their attributes, special attribute types and business rules were defined for the Sangarilla Hotel Database System. Chapter 3 transforms this analysis into the logical and physical database designs.

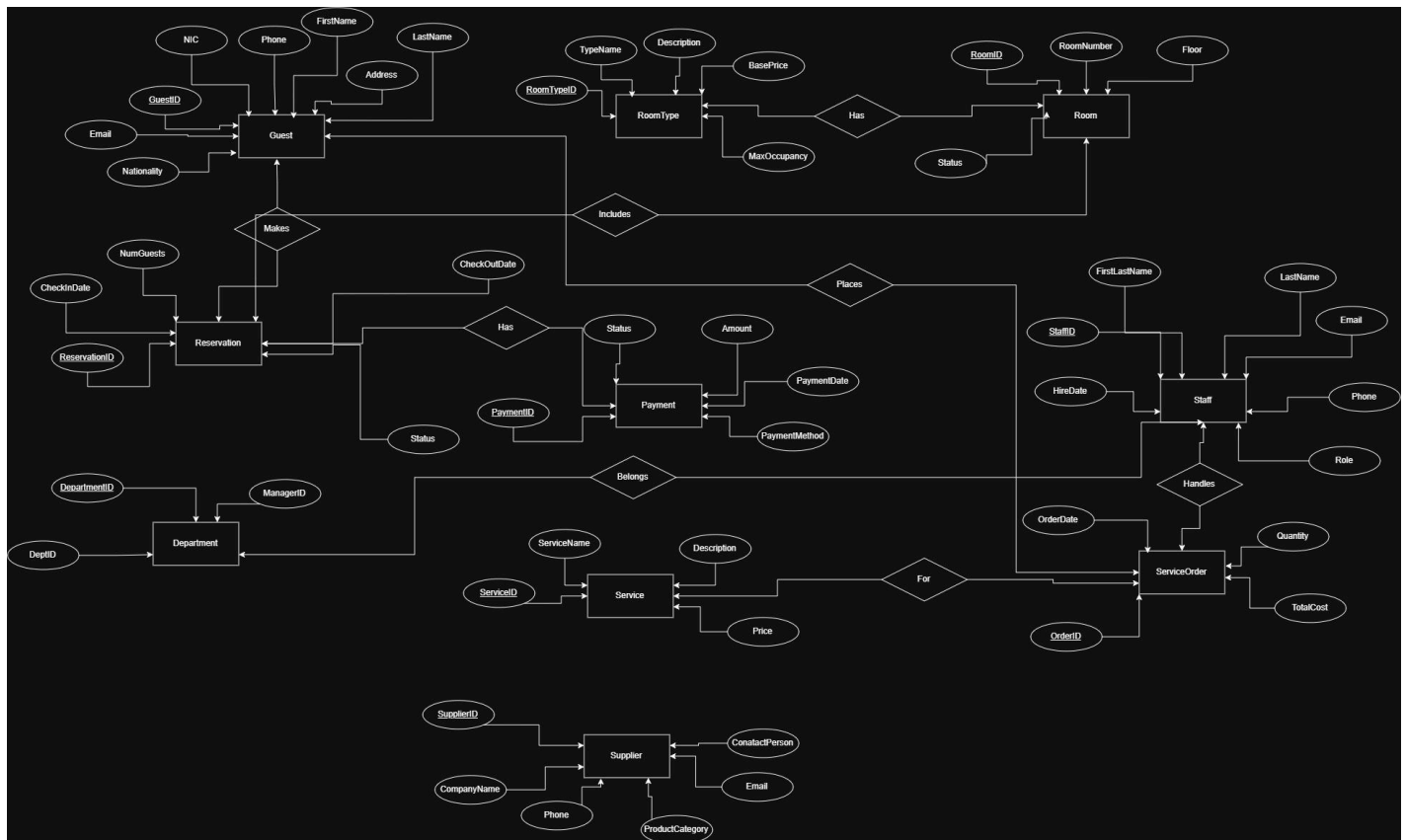
Chapter 3: Design

3.1 Introduction

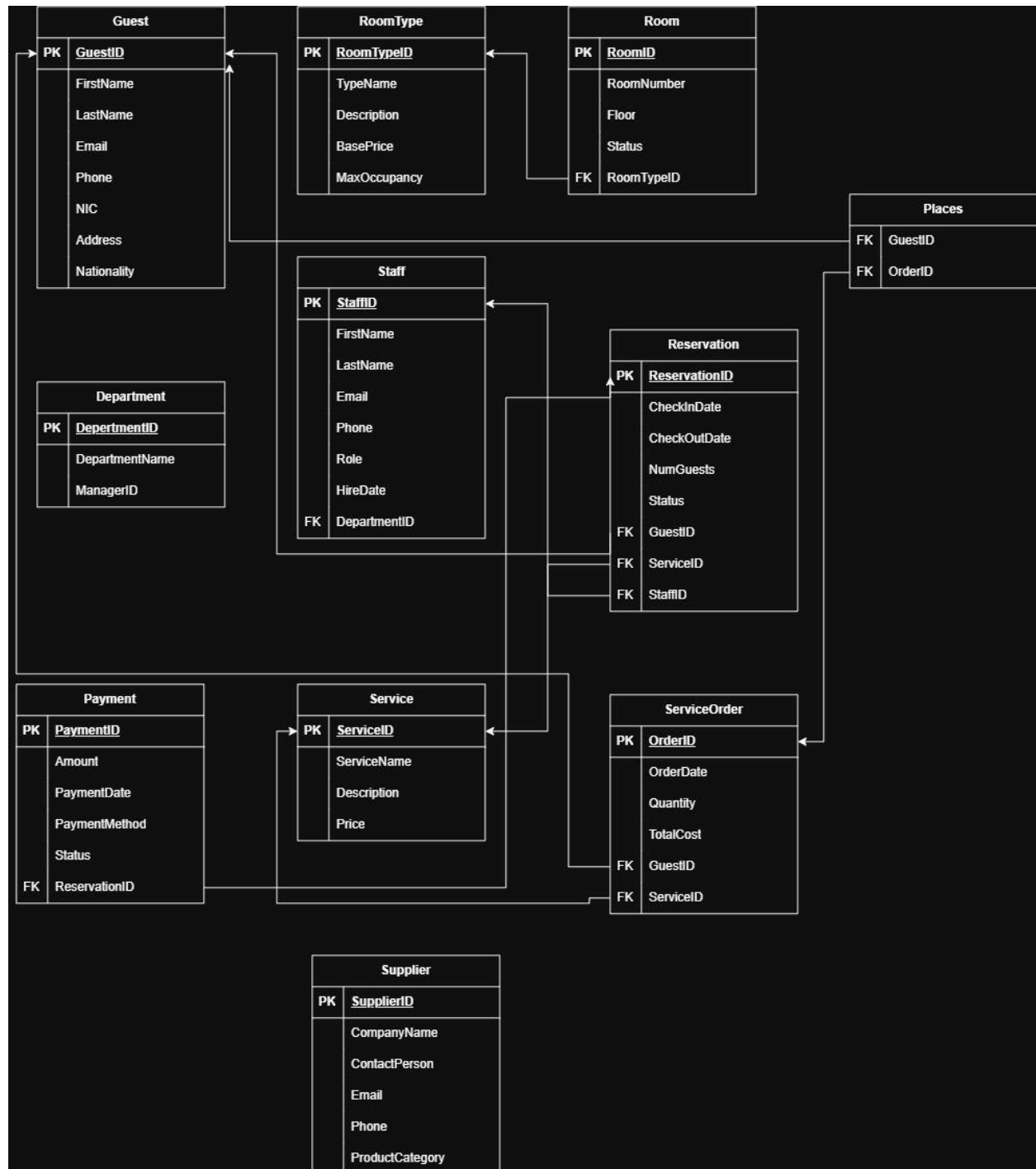
This chapter talks about the three levels of database design that are needed for the coursework: the design (ER diagram) the logical design (relational schema) and the physical design (SQL Server implementation and security), for the Sangarilla Hotel Database System.

3.2 Conceptual Design (ER Diagram)

Figure 3.1 shows the planned ER diagram for the Sangarilla Hotel Database System. Rectangles represent entities, diamonds represent relationships, a double line indicates participation and a single line indicates partial participation. The final ER diagram will be drawn on a Bristol board with all required attributes.



3.2 Logical (Relational Schema Mapping)



Chapter 4: Conclusion

4.1 Summary

This proposal explains the design of the Sangarilla Hotel Database System for the Data Management 1 Coursework. The requirements were studied and ten entities were found in an ER diagram, with cardinality and participation was. The relational schema was made and cleaned up to Third Normal Form. Each table was also described with its columns, data types and sizes. The SQL Server implementation FAQ queries and five-level security design finish the proposed solution.

4.2 Advantages of the Database

- Reduces data redundancy and inconsistency by organizing data in a way.
- Keeps data accurate by using keys foreign keys and check constraints.
- Links guests, bookings, payments and rooms so that the entire booking process can be followed.
- Helps answer business questions fast, by using SQL queries and joining tables.
- Keeps data safe by having user levels and permissions.
- Makes it easy to add guests, rooms, bookings and services as the hotel grows.

4.3 Limitations and Future Enhancements

- The current scope covers the Sangarilla Hotel database. In the future a web interface and payment integration can be added.
 - Additional tables such, as discounts, customer feedback and inventory purchases can be added in versions.
 - The system can be expanded as the hotel's requirements grow.
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